

L05. The Linear Quadratic Regulator (LQR), continued

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1 Introduction

We continue our exploration from last lecture on the Linear Quadratic Control problem. From the previous lecture, recall the finite-horizon LQR problem:

$$x_{t+1} = A_t x_t + B_t u_t + w_t, \quad t = 0, \dots, T-1,$$

where w_t is i.i.d. zero-mean noise. The cost at each step and the terminal cost are

$$c_t(x_t, u_t) = x_t^\top Q_t x_t + u_t^\top R_t u_t, \quad c_T(x_T) = x_T^\top Q_T x_T.$$

Our objective is to choose a policy π that minimizes

$$\mathbb{E} \left[c_T(x_T) + \sum_{t=0}^{T-1} c_t(x_t, u_t) \right].$$

We assume Q_t is positive semidefinite and R_t is positive definite.

Last lecture, we found that the optimal control is linear in the state and that its quadratic value function can be computed backward using the Riccati equation. For a stationary system, where $A_t = A$, $B_t = B$, $Q_t = Q$, and $R_t = R$, this recursion becomes

$$\begin{aligned} K_T &= Q_T, \\ G_t &= R + B^\top K_{t+1} B, \\ K_t &= A^\top (K_{t+1} - K_{t+1} B G_t^{-1} B^\top K_{t+1}) A + Q, \\ u_t^*(x_t) &= -G_t^{-1} B^\top K_{t+1} A x_t. \end{aligned}$$

In the finite-horizon problem, we start from $K_T = Q_T$ and compute the Riccati recursion backward for the remaining time steps. In this lecture, we ask the question: do these finite-horizon solutions converge to a stabilizing stationary controller as $T \rightarrow \infty$?

2 Reachability

To answer these questions, we first examine what our inputs can do in our system. We check *reachability* using the dynamics without the noise term, since random disturbances cannot be chosen as controls:

$$x_{t+1} = A x_t + B u_t.$$

The system is reachable if, for any initial state x_0 and any desired final state x_f , we can choose a sequence of inputs $u_0, u_1, \dots, u_{N-1}$ that takes the system from x_0 to x_f in a finite number of steps N .

We check reachability using the matrix

$$\mathcal{R} = [B \quad AB \quad \cdots \quad A^{n-1}B].$$

The system is reachable exactly when

$$\text{rank}(\mathcal{R}) = n.$$

Intuitively, the columns of $B, AB, \dots, A^{n-1}B$ describe the directions in which our inputs can affect the state, either immediately or after several time steps. Since the state $x_t \in \mathbb{R}^n$ has n coordinates, these columns must span $\mathbb{R}^n$ if we want to reach every possible final state. That is exactly what $\text{rank}(\mathcal{R}) = n$ says.

We can express the same condition using the infinite-step reachability Gramian, when the sum converges:

$$P = \sum_{i=0}^{\infty} A^i B B^\top (A^i)^\top.$$

P records the directions in state space that the inputs can affect: $x^\top P x$ is zero if the inputs cannot affect the direction x , and positive if they can. Thus, the system is reachable iff P is positive definite.

This definition gives us a cleaner way to write out the reachability condition. Separating out the $i = 0$ term gives

$$\begin{aligned} P &= B B^\top + \sum_{i=1}^{\infty} A^i B B^\top (A^i)^\top \\ &= B B^\top + A \left(\sum_{j=0}^{\infty} A^j B B^\top (A^j)^\top \right) A^\top \\ &= B B^\top + A P A^\top. \end{aligned}$$

Thus $P - A P A^\top = B B^\top$ which we provides another way of viewing stability.

Example

For example,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Here $AB = (b_2, 0, 0)^\top$ and $A^2B = 0$, so

$$\mathcal{R} = [B \quad AB \quad A^2B] = \begin{bmatrix} b_1 & b_2 & 0 \\ b_2 & 0 & 0 \\ b_3 & 0 & 0 \end{bmatrix}.$$

The third column is zero, so $\text{rank}(\mathcal{R}) \leq 2 < 3$. Thus, no choice of the values b_1, b_2, b_3 makes this system reachable: the inputs cannot move the state independently in all three directions.

3 Observability

We can also examine what the system's outputs tell us about its state. We check *observability* using the dynamics without inputs or noise:

$$x_{t+1} = A x_t, \quad y_t = C x_t.$$

The system is observable if no nonzero initial state x_0 produces an output that is identically zero. In other words, we can distinguish any initial state from the zero state by looking at its outputs over time.

We check observability using the matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}.$$

The system is observable exactly when

$$\text{rank}(\mathcal{O}) = n.$$

Again, intuitively, the rows of $C, CA, \dots, CA^{n-1}$ describe which parts of the initial state appear in the output, either immediately or after several time steps. Since $x_0 \in \mathbb{R}^n$, the matrix must have rank n to distinguish every initial state.

We can express the same condition using the infinite-step observability Gramian, when the sum converges:

$$Q = \sum_{i=0}^{\infty} (A^i)^\top C^\top C A^i.$$

Q records how the initial state appears in the outputs: $x^\top Q x$ is zero if the direction x produces zero output at every step, and positive if it appears in at least one output. Thus, the system is observable iff Q is positive definite.

Separating out the $i = 0$ term gives

$$\begin{aligned} Q &= C^\top C + \sum_{i=1}^{\infty} (A^i)^\top C^\top C A^i \\ &= C^\top C + A^\top \left(\sum_{j=0}^{\infty} (A^j)^\top C^\top C A^j \right) A \\ &= C^\top C + A^\top Q A. \end{aligned}$$

Thus $Q - A^\top Q A = C^\top C$, which we provides another way of viewing observability.

Example

For example, the slides consider

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad C = [1 \quad 0 \quad 0].$$

Here $C = CA = CA^2 = [1 \quad 0 \quad 0]$, so

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

We have $\text{rank}(\mathcal{O}) = 1 < 3$, or equivalently $\ker(\mathcal{O}) \neq \{0\}$. The output reveals only the first coordinate, so the system is not observable.

These properties will play different roles in the infinite-horizon LQR argument. Reachability gives us a stabilizing controller, which provides a finite upper bound on the optimal cost. If we write the state cost matrix as $Q = C^\top C$, observability ensures that no nonzero state trajectory can remain invisible to that cost. We use both ideas below to study the limiting optimal controller.

4 Stability

Consider a pendulum at equilibrium. It gives three examples of stability in systems. Balanced upright, it is unstable: a small disturbance makes it fall away from the equilibrium. Hanging downward without friction, it is Lyapunov stable: after a small disturbance it swings nearby, but does not settle. Hanging downward with friction, it is asymptotically stable: its swings diminish until it returns to rest at the equilibrium.

4.1 Defining stability

To make this distinction precise, consider

$$x_{k+1} = f(x_k), \quad f(0) = 0,$$

where $x = 0$ is an equilibrium. The equilibrium is *Lyapunov stable* if every sufficiently small change in the initial state stays small: for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\|x_0\| < \delta \implies \|x_k\| < \varepsilon \quad \text{for all } k \geq 0.$$

The state does not have to return to zero. It is *asymptotically stable* if it is Lyapunov stable and, for initial states sufficiently close to zero,

$$x_k \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

One way to establish stability is to find a function $V(x)$ that is zero at the equilibrium and positive at nearby nonzero states. We examine its change along a trajectory:

$$\Delta V(x) = V(f(x)) - V(x).$$

If $\Delta V(x) \leq 0$ near the equilibrium, then V does not increase, providing a Lyapunov stability argument. If $\Delta V(x) < 0$ at every nearby nonzero state, its value strictly decreases, providing an asymptotic stability argument.

4.2 Connecting stability to LQR

We now return to LQR with two questions: as the horizon grows, do the finite-horizon optimal controllers approach a stationary controller? If they do, does that controller make the closed-loop system stable?

For a linear system $x_{k+1} = Ax_k$, a natural Lyapunov function is $V(x) = x^\top Hx$, where $H \succ 0$. Along the trajectory,

$$\Delta V(x) = x^\top (A^\top H A - H)x.$$

If we can find $H \succ 0$ such that $A^\top H A - H \prec 0$, the origin is asymptotically stable. For this discrete-time linear system, this occurs exactly when every eigenvalue of A has magnitude strictly less than 1.

Now return to LQR. Under a stationary feedback law $u_k = Lx_k$, the dynamics without noise become

$$x_{k+1} = Ax_k + Bu_k = (A + BL)x_k, \quad x_k = (A + BL)^k x_0.$$

Here $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, and $L \in \mathbb{R}^{m \times n}$. To make the state converge to zero, we want every eigenvalue of the closed-loop matrix $A + BL$ to lie strictly inside the unit circle.

Can we always choose L to make this happen? **Reachability** of (A, B) guarantees that some stabilizing feedback gain exists. This matters when we take the LQR horizon to infinity: it gives us a controller whose cost we can use to bound the optimal cost. But it does not yet tell us whether the *optimal* feedback law is stable. We first show that the Riccati matrices converge by proving boundedness and monotonicity, then examine the stability of the limiting optimal controller.

4.3 Boundedness and Monotonicity

For this convergence argument, we study the quadratic cost matrix under the noise-free dynamics and take the terminal cost to be zero. We have a Riccati recursion for each finite horizon. To ask whether it converges, let K_k denote the quadratic value matrix for a problem with k steps remaining and zero terminal cost. Then $K_0 = 0$, and the Riccati recursion computes K_{k+1} from K_k . We want to show that

$$K_k \longrightarrow K \quad \text{as } k \rightarrow \infty.$$

We do this by showing that the matrices are bounded and monotone.

4.3.1 Boundedness

Reachability guarantees that we can choose a feedback gain L for which $A + BL$ is stable. If we apply $u_t = Lx_t$, the noise-free state is $x_t = (A + BL)^t x_0$. Because these states decay toward zero, the total cost of this particular policy is finite and gives a bound of the form

$$\sum_{t=0}^{\infty} (x_t^\top Q x_t + u_t^\top R u_t) \leq c \|x_0\|^2$$

for some constant c . The optimal k -step cost cannot exceed the cost of this policy. Thus

$$x_0^\top K_k x_0 \leq c \|x_0\|^2 \quad \text{for every } k.$$

4.3.2 Monotonicity

Adding one more step cannot reduce the optimal cost when each stage cost is nonnegative and the terminal cost is zero. Given any strategy for $k + 1$ steps, we can ignore its last step; its cost over the first k steps is no greater than its full cost, and the optimal k -step cost is no greater than that first part. Hence

$$x^\top K_k x \leq x^\top K_{k+1} x \quad \text{for every } x,$$

or equivalently, $K_k \preceq K_{k+1}$. The matrices are therefore increasing but bounded above, so they converge to a limit K . We can then use this limit to study the stationary optimal feedback law.

4.4 Stability under optimal control

Taking the limit in the Riccati recursion gives the algebraic Riccati equation

$$K = Q + A^\top K A - A^\top K B (R + B^\top K B)^{-1} B^\top K A.$$

It gives a candidate stationary optimal feedback law

$$L = -(R + B^\top KB)^{-1} B^\top KA, \quad u_k = Lx_k.$$

Under this law, the noise-free closed-loop dynamics are $x_{k+1} = A_{\text{cl}}x_k$, where $A_{\text{cl}} = A + BL$.

We still need to show that the limiting law stabilizes the system. A natural Lyapunov function candidate is the limiting value function $V(x) = x^\top Kx$. The Riccati equation gives

$$\begin{aligned} V(x_{k+1}) - V(x_k) &= x_k^\top (A_{\text{cl}}^\top KA_{\text{cl}} - K)x_k \\ &= -(x_k^\top Qx_k + u_k^\top Ru_k) \leq 0. \end{aligned}$$

Thus V decreases by exactly the cost paid at each step. If $Q = C^\top C$ and (A, C) is observable, no nonzero trajectory can keep paying zero cost: zero input cost forces $u_k = 0$, while zero state cost forces $Cx_k = 0$ at every step. **Observability** then forces the initial state to be zero. Consequently, the limiting closed-loop system is asymptotically stable, so $V(x_k) \rightarrow 0$.

Summing the displayed identity over time now shows that this stationary law has total cost $V(x_0) = x_0^\top Kx_0$. Since K is the limit of the finite-horizon *optimal* costs, no policy can have a lower infinite-horizon cost. The limiting feedback law is therefore optimal and stabilizing.

Next lecture, we will discuss how this convergence argument in LQR holds in more general cases for *value iteration*.